SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

**SLOT:** A

**COURSE:** Fundamentals of Data Science

**COURSE CODE:** DSA 0404

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**1.Find mean, median, mode and range for: 12, 15, 18, 15, 20, 22, 15, 25.**

**Given data:**

12,15,18,15,20,22,15,25

**Mean**

**Formula:**

$$\bar{x} = \sum x/n$$

$$\sum x = 12 + 15 + 18 + 15 + 20 + 22 + 15 + 25 = 142$$

$$\bar{x} = \frac{142}{8} = 17.75$$

**Median**

Arrange the data:

12,15,15,15,18,20,22,25

Since n=8:

Median=$4^{th}$+$5^{th}$/2

=15+18/2=16.5

**Mode**

15 occurs most frequently.

Mode=15

**Range**

**Formula:**

Range=Maximum−Minimum

=25−12=13

**Answer**

- Mean = 17.75
- Median = 16.5
- Mode = 15
- Range = 13

**2. Construct a frequency distribution for: 2, 3, 4, 2, 5, 3, 4, 2, 6, 5, 3, 4.**

**Given data:**

$$2,3,4,2,5,3,4,2,6,5,3,4$$

| Value | Frequency |
|---|---|
| 2 | 3 |
| 3 | 3 |
| 4 | 3 |
| 5 | 2 |
| 6 | 1 |
| **Total** | **12** |

Therefore, the frequency distribution is:

$$\boxed{2{:}3,\ \ 3{:}3,\ \ 4{:}3,\ 5{:}2,\ \ 6{:}1}$$

**3. Find Q1, Q2, Q3 and IQR for: 5, 8, 10, 12, 15, 18, 20, 22, 25, 30.**

**Given data:**

$$5,8,10,12,15,18,20,22,25,30$$

The data is already arranged in ascending order.

**Q2 (Median)**

There are 10 values:

$$Q2=15+18/2=16.5$$

**Q1**

Lower half:

$$5,8,10,12,15$$

Middle value:

$$Q1=10$$

**Q3**

Upper half:

$$18,20,22,25,30$$

Middle value:

$$Q3=22$$

**IQR**

**Formula:**

$$IQR=Q3-Q1$$

$$IQR=22-10=12$$

**Final Answer:**

- Q1 = 10
- Q2 = 16.5
- Q3 = 22
- IQR = 12

**4. Detect outliers using the IQR method for: 10, 12, 13, 15, 16, 17, 18, 20, 45.**

**Given data:**

$$10,12,13,15,16,17,18,20,45$$

**Q1 and Q3**

Lower half:

$$10,12,13,15$$

$$Q1=12+13/2=12.5$$

Upper half:

$$17,18,20,45$$

$$Q3=18+20/2=19$$

**IQR**

$$IQR=Q3-Q1$$

$$=19-12.5=6.5$$

**Outlier Formula**

$$\text{Lower Fence} = Q1 - 1.5(IQR)$$

$= 12.5 - 1.5(6.5)$

$= 12.5 - 9.75 = 2.75$

$$\text{Upper Fence} = Q3 + 1.5(IQR)$$

$$= 19 + 9.75 = 28.75$$

Any value below 2.75 or above 28.75 is an outlier.

Since:

$$45 > 28.75$$

$$\boxed{45 \text{ is an outlier}}$$

**Q5. Determine the skewness (Asymmetry) of: 5, 6, 7, 7, 8, 8, 9, 20.**

Dataset: 5, 6, 7, 7, 8, 8, 9, 20 (n = 8)

**Mean**

$$Mean = \Sigma x / n$$

$$\text{Sum} = 5+6+7+7+8+8+9+20 = 70$$

$$\text{Mean} = 70/8 = 8.75$$

**Median**

Sorted data:

$$5,6,7,7,8,8,9,20$$

$n=8$ (even)

$$\text{Median} = \text{average of 4th \& 5th values}$$

$$= (7+8)/2 = 7.5$$

**Standard Deviation (population)**

$$\sigma = \sqrt{\Sigma(x - \bar{x})^2 / n}$$

Squared deviations from mean 8.75:

$$14.06, 7.56, 3.06, 3.06, 0.56, 0.56, 0.06, 126.56$$

Sum of squared deviations

$$\sigma = \sqrt{(155.5 / 8)} = \sqrt{19.44} = 4.41$$

**Pearson's Coefficient of Skewness**

$$Skewness = 3 \times (Mean - Median) / \sigma$$

Skewness:

$$3 \times (8.75 - 7.5) / 4.41$$

$$= 3 \times 1.25 / 4.41$$

$$= 3.75 / 4.41 \approx 0.85$$

**Interpretation**

- Mean (8.75) > Median (7.5) → distribution is right-skewed (positively skewed)
- Skewness coefficient ≈ +0.85 confirms moderate positive skew
- The value 20 is a high extreme value pulling the mean to the right

**Q6. A uniformly distributed variable ranges from 10 to 50. Find P(20<X<35).**

X is uniformly distributed on the interval [10, 50], i.e. $X \sim U(10, 50)$

**Formula**

*For* $X \sim U(a, b)$*:* $f(x) = 1 / (b - a)$, $P(c < X < d) = (d - c) / (b - a)$

Here a = 10, b = 50

So, total width = b − a = 50 − 10 = 40

Probability density f(x) = 1/40 for $10 \le x \le 50$

Required interval width = 35 − 20 = 15

$$P(20 < X < 35) = 15 / 40 = 0.375$$

**Answer:** $P (20 < X < 35) = 0.375$ (37.5%)

**Q7. A normal distribution has mean 100 and standard deviation 15. Find the Z-score for X=130.**

Given: Mean ($\mu$) = 100, Standard Deviation ($\sigma$) = 15

**Formula**

$$Z = (X - \mu) / \sigma$$

$$Z = (130 - 100) / 15$$

$$= 30 / 15$$

$$= 2$$

**Q8. Find P(X < 130) for the Same Normal Distribution**

Using $\mu = 100$, $\sigma = 15$, and the Z-score computed in Q7 ($Z = 2$)

**Formula**

$$P(X < x) = P(Z < (x - \mu)/\sigma) = \Phi(Z)$$

$$Z = (130 - 100)/15 = 2 \text{ (from Q7)}$$

Look up $\Phi(2)$ in the standard normal (Z) table

$$\Phi(2) = 0.9772$$

**Answer:** $P(X < 130) = 0.9772$ ($\approx$ 97.72%)

**Q9. Given a dataset with one extreme value, compare the mean and median before and after outlier treatment.**

Sample dataset with an extreme value: 10, 12, 13, 14, 15, 100

**Before Outlier Treatment**

**Mean:**

$$Mean = \Sigma x/n$$

Sum = 10+12+13+14+15+100 = 164, n = 6

$$\text{Mean} = 164/6 = 27.33$$

***Median:***

*Median = middle value(s) of sorted data*

Sorted:

$$10,12,13,14,15,100$$

$$\text{Median} = \text{avg}(13,14) = 13.5$$

**After Removing the Outlier (100)**

Treated dataset: 10, 12, 13, 14, 15 (n = 5)

**Mean:**

$$\text{Sum} = 10+12+13+14+15 = 64$$

$$\text{Mean} = 64/5 = 12.8$$

**Median:**

$$\text{Median (middle of 5 values)} = 13$$

**Comparison Table**

| Measure | Before Treatment | After Treatment | Change |
|---|---|---|---|
| Mean | 27.33 | 12.80 | Drops sharply by 14.53 |
| Median | 13.5 | 13.0 | Changes only slightly by 0.5 |

**Conclusion**

- The mean is highly sensitive to extreme values (pulled strongly toward the outlier)
- The median is robust/resistant to outliers and barely changes
- For skewed data or data with outliers, the median is a more reliable measure of central tendency

**Q10. Perform a complete EDA on a small student-mark dataset and report central tendency, dispersion, distribution, outliers and skewness.**

Dataset (marks out of 100, sorted): 45, 50, 55, 60, 62, 65, 70, 75, 80, 95 (n = 10)

**1. Central Tendency**

**Mean:**

$$Mean = \Sigma x/n$$

Sum = 45+50+55+60+62+65+70+75+80+95 = 657

Mean = 657/10 = 65.7

**Medain:**

Median = average of 5th & 6th values

= (62+65)/2 = 63.5

**Mode:**

Mode = none (all values are unique / no repeats)

**2. Dispersion**

$$Range = Max - Min$$

$$= 95 - 45 = 50$$

Squared deviations from mean (65.7):

428.5, 246.5, 114.5, 32.5, 13.7, 0.5, 18.5, 86.5, 204.5, 858.5

$$\sigma = \sqrt{(\Sigma(x-\bar{x})^2/n)}$$

Sum of squares = 2004.1

Variance ($\sigma^2$) = 2004.1/10 = 200.41

**Standard deviation σ** = $\sqrt{200.41} \approx 14.16$

**3. Quartiles & IQR**

Lower half:

45,50,55,60,62

Q1 = 55 (middle value)

Upper half:

$$65,70,75,80,95$$

$$Q3 = 75 \text{ (middle value)}$$

**IQR** = Q3 − Q1 = 75 − 55 = 20

**4. Outlier Check (IQR method)**

Lower Bound = Q1 − 1.5×IQR = 55 − 30 = 25

Upper Bound = Q3 + 1.5×IQR = 75 + 30 = 105

All values lie within [25, 105] → no outliers detected (95 is high but still within bound)

**5. Skewness**

$$Skewness = 3 \times (Mean - Median) / \sigma$$

Skewness:

$$3 \times (65.7 - 63.5) / 14.16$$

$$= 3 \times 2.2 / 14.16 \approx 0.47$$

Mean (65.7) slightly greater than Median (63.5) → mild right (positive) skew

**Summary Table**

| Metric | Value |
|---|---|
| Mean | 65.7 |
| Median | 63.5 |
| Mode | None |
| Range | 50 |
| Q1 | 55 |
| Q3 | 75 |

| | |
|---|---|
| IQR | 20 |
| Std. Deviation | ≈14.16 |
| Outliers | None |
| Skewness | ≈+0.47 (mild positive) |

**Overall Interpretation**

- Students' marks are centered around 64–66 with moderate spread ($\sigma \approx 14.16$)
- No mode exists — every student scored a unique mark
- No true outliers, though the top score (95) pulls the mean slightly above the median
- The mild positive skew suggests a few higher scorers stretch the right tail of the distribution, while most students cluster in the mid-range

**Answer:** The dataset is approximately symmetric with a mild positive skew, moderate spread, and no statistical outliers.